

Iberoamerican Mathematical Olympiad 2026

Friday, September 25, 2026

Problems from the 41st Iberoamerican Mathematical Olympiad (Buenos Aires, 2026), translated from Spanish.

Toolkit

- ◇ **Partial sums and residues.** For a sequence (a_n) , write

$$S_n = a_1 + \cdots + a_n$$

for its partial sums. Modulo m , the recurrence $S_n = S_{n-1} + a_n$ has only m possible states.

- ◇ **Term from sums and averages.** Note that

$$S_n = S_{n-1} + a_n \Rightarrow S_n - S_{n-1} = a_n.$$

If $n \mid S_n$, call $b_n = S_n/n$. Then

$$a_n = nb_n - (n-1)b_{n-1} = b_{n-1} + n(b_n - b_{n-1}) \quad (n \geq 2).$$

- ◇ **Polynomial divisibility.** If $p \in \mathbb{Z}[x]$ and $u, v \in \mathbb{Z}$, then

$$u - v \mid p(u) - p(v).$$

Also, if $1 = d_1 < \cdots < d_k = n$ are all positive divisors of n , then $d_i d_{k+1-i} = n$.

Definition. A partial order on a set A is a binary relation $\leq$ which is

- (a) **Reflexive.** For any $a \in A$, $a \leq a$.
- (b) **Antisymmetric.** For $a, b \in A$, $a \leq b$ and $b \leq a$ imply $a = b$.
- (c) **Transitive.** For $a, b, c \in A$, if $a \leq b$ and $b \leq c$, then $a \leq c$.

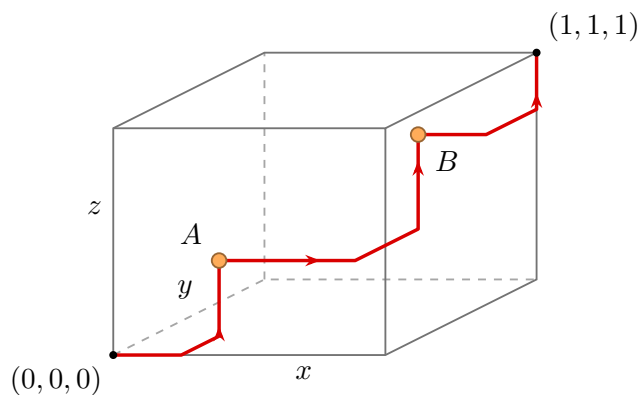
Example. For a set A , its power set along with the containment relation form a partial order $(\mathcal{P}(A), \subseteq)$.

Definition. In a poset (partially ordered set), a chain is totally ordered subset. This is, every element is comparable.

On the other hand, an antichain is a set of mutually incomparable elements.

Theorem (Dilworth). *In a finite poset, the maximum size of an antichain equals the minimum number of chains needed to cover all elements. This number is called the width of the partial order.*

Help for Problem 3



The red path collects both alfajores: $A = (\frac{1}{4}, \frac{1}{4}, \frac{1}{3})$ and $B = (\frac{3}{4}, \frac{2}{3}, \frac{3}{4})$. Every segment is parallel to a cube edge, and no coordinate decreases.



Problems: Day 1

Exercise 1 (Iberoamerican Olympiad 2026, Problem 1). Let $a_1, a_2, \dots, a_{1000}$ be a permutation of the integers $1, 2, \dots, 1000$. Determine all possible values of a_1 for which, regardless of how the remaining integers are ordered, there is an integer k , $1 \leq k \leq 1000$, such that

$$a_1 + a_2 + \dots + a_k$$

is divisible by 3.

Exercise 2 (Iberoamerican Olympiad 2026, Problem 2). Does there exist a bijection $f : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$ satisfying all of the following conditions?

- (i) For every positive integer n , the sum $f(1) + f(2) + \dots + f(n)$ is divisible by n .
- (ii) For every positive integer n ,

$$f(1) + f(2) + \dots + f(n) \leq n^2.$$

- (iii) For infinitely many positive integers n ,

$$f(1) + f(2) + \dots + f(n) = n^2.$$

Note: Bijectivity means that each positive integer appears exactly once in the list $f(1), f(2), \dots$.

Exercise 3 (Iberoamerican Olympiad 2026, Problem 3). Mafalda dreams of taking an interstellar trip with her friends. On the journey, she discovers a cube-shaped room containing 2026 motionless *alfajores* floating inside it. She chooses a vertex of the cube and places all her accompanying friends there.

Each friend must travel to the opposite vertex, moving only in directions parallel to the edges of the cube and always away from the starting vertex, collecting every alfajor encountered along the way.

What is the minimum number of friends Mafalda must bring to guarantee that they can collect all the alfajores, regardless of their arrangement?

Note: Treat the friends and the alfajores as points in space.

Problems: Day 2

Exercise 4 (Iberoamerican Olympiad 2026, Problem 4). A sequence $a_1, a_2, \dots$ of positive integers is called *porteña* if it satisfies the following conditions:

- (i) $a_1 < a_2 < a_3 < \dots$.
- (ii) $2a_n - n$ is a perfect square for every $n \geq 1$.
- (iii) $a_{n-1} + a_n \leq n^2$ for every $n \geq 2$.

Determine all *porteña* sequences.

Exercise 5 (Iberoamerican Olympiad 2026, Problem 5). Let ABC be an acute triangle with $AB > AC$, inscribed in a circle Γ . A point X , distinct from A and C , varies along the arc AC of Γ that does not contain B . The lines AX and BC meet at P .

The line through P parallel to AB meets the tangent to Γ at B at D . The line through P parallel to AC meets the tangent to Γ at C at E .

Prove that the circumcenter of triangle XDE lies on a fixed line.

Note: The circumcenter of a triangle is the center of the circle through its three vertices.

Exercise 6 (Iberoamerican Olympiad 2026, Problem 6). Determine all integers $n > 1$ for which there exist a permutation $d_1, d_2, \dots, d_k$ of all the positive divisors of n and a polynomial $P(x) \in \mathbb{Z}[x]$ such that

$$P(d_j) = d_{j+1} \quad (1 \leq j \leq k-1).$$